

EMRAH FAJKOVIC

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Summary

Senior Software Engineer with **11+ years** of experience in **full-stack development, cloud infrastructure, and AI/ML solutions**. Proven expertise in designing scalable microservices, deploying AI-driven applications, and optimizing MLOps workflows on **AWS, GCP, and Azure**. Adept at leveraging **Python, Java, and JavaScript** (React/Angular) to build high-performance systems. Strong background in **Generative AI (LLMs, GPT, BERT), data pipelines (Kafka, Airflow), and DevOps (Docker, Kubernetes)**. Passionate about delivering innovative, scalable solutions in **Agile** environments.

Skills

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| <ul style="list-style-type: none">• Python• JavaScript• Java• TypeScript• SQL• C/C++• Scala | <ul style="list-style-type: none">• Frontend (React, Angular, Vue.js)• Backend (Django, Flask, Spring Boot, FastAPI)• Cloud Platforms• DevOps & CI/CD• ML & AI• Agile Environment• Leadership |
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Experience

Senior Software Engineer

Ciklum – United States | 04/2022 – Present

- Developed and deployed advanced **machine learning models** leveraging structured and unstructured data with **Python** and **Java**, enhancing customer experience and providing real-time predictions on user needs.
- Built and optimized **large-scale data processing pipelines** using PySpark and Scala on Spark clusters, handling vast datasets for efficient transformation and model training.
- Led the design and deployment of **generative AI models**, including transformer architectures like GPT and BERT, optimized for diverse language processing tasks and fine-tuned for specific industries.
- Utilized **NVIDIA tools** such as TensorRT and CUDA to optimize **deep learning models**, ensuring high-performance inference and reduced latency in GPU environments.
- Employed **MLOps best practices**, integrating CI/CD pipelines and model versioning to streamline the machine learning lifecycle and ensure continuous improvement.
- Designed and implemented scalable, end-to-end machine learning workflows on **Google Cloud’s Vertex AI**, integrating services like BigQuery, Dataflow, and AlloyDB for robust data engineering and model orchestration.
- Developed **user-friendly front-end interfaces** using React and JavaScript to visualize AI-driven insights, enabling users to interact with data and AI models seamlessly.
- Pioneered the exploration of **Large Language Models (LLMs)** and Generative AI techniques for innovative content recommendation and personalization use cases, contributing to cutting-edge AI-driven solutions.
- Built data pipelines and batch processing solutions using **Apache Kafka and Airflow**, ensuring reliable, low-latency data flows between systems for real-time AI applications.
- Deployed machine learning models in production using **Docker** and **Kubernetes** for containerization, achieving scalable and efficient model delivery across cloud environments.
- Leveraged **Snowflake’s Snowpark** and advanced SQL techniques to manage large-scale data analytics and preprocessing, facilitating seamless integration with machine learning workflows.

Environment: Python, Java, PySpark, Scala, TensorFlow, PyTorch, NVIDIA (TensorRT, CUDA), MLOps, LLMs, Generative AI, Kubernetes, Docker, Apache Kafka, Apache Airflow, React, JavaScript, GCP, Vertex AI, BigQuery, Dataflow, AlloyDB, Snowflake, PostgreSQL, MySQL, Transformer Models, Scikit-learn, SQL

Senior Full Stack Developer

Bank of America – United States | 11/2019 – 03/2022

- Built and deployed **RESTful APIs using FastAPI, Django, and Java** to manage user data and automate tasks, significantly improving data accessibility and streamlining workflows in a fintech environment.
- Developed and maintained **financial instrument logic** using in-house client libraries and migrated solutions to newer **Python** and **Java** versions, ensuring compatibility and performance improvements across services.
- Utilized in-house Quartz on-prem solution, leveraging its custom database job scheduler and IDE, alongside Python and Java libraries, to contribute to financial instrument development and updates.
- Developed **monitoring** and **notification tools** in Python and Java for proactive issue **identification** and **resolution**, enhancing system reliability and improving user experience.
- Built dynamic, user-centric **front-end applications using React and Vue.js**, creating intuitive interfaces for financial data access and user engagement.
- Containerized applications using **Docker** to ensure scalability and streamlined deployment across development and production environments.
- Performed **troubleshooting, debugging**, and deployment of Python and Java bug fixes, minimizing downtime and improving system stability.
- Designed and optimized **SQL Server** and **Oracle** database schemas to handle complex financial data structures, ensuring data accuracy and integrity.
- Managed and utilized **AWS services**, including Lambda, S3, DynamoDB, and Kinesis, to implement efficient data processing and storage solutions, optimizing infrastructure costs.
- Developed **data extraction** and **ETL processes** using Python, enhancing data accuracy and availability for critical financial analysis.
- Established and managed CI/CD pipelines using **AWS CodePipeline** and **Jenkins** to automate testing, deployment, and reduce release cycle times.
- Employed **Agile** and **Scrum** methodologies, delivering high-quality software solutions that met project deadlines and business requirements in the fintech domain.

Environment: Python, Java, FastAPI, Django, Quartz, Spring Boot, React, Vue.js, JavaScript, SQL Server, Oracle, Docker, Jenkins, AWS CodePipeline, AWS (Lambda, S3, DynamoDB, and Kinesis), Pandas, ETL Pipelines, Agile/Scrum, SDLC

Full Stack Developer

Luma Health – United States | 06/2016 – 10/2019

- Architected backend services with **Python and Django, ensuring HIPAA-compliant data handling and FHIR integration** for healthcare applications.
- Built and deployed **RESTful APIs** using Django and FastAPI to facilitate seamless data exchange, including integration with third-party healthcare systems following FHIR protocols.
- Designed and optimized **PostgreSQL and MongoDB** schemas to support complex healthcare workflows, with a focus on **FHIR data structures** to enhance interoperability and data consistency.
- Developed dynamic, responsive **front-end applications using Angular/React**, creating intuitive interfaces for healthcare professionals and patients to improve user engagement.
- **Collaborated closely with front-end teams** to integrate Angular/React components with Django backend services, delivering cohesive and scalable full-stack applications.
- Implemented **data processing** and **transformation pipelines** using **Python**, enhancing data quality and availability for healthcare analytics and reporting.
- Established **CI/CD pipelines** with **Jenkins, Docker, and Git** to automate testing and deployment processes, reducing time-to-release for critical features.
- Leveraged **AWS services**, including S3 and Lambda, for scalable, cost-efficient solutions to store and process healthcare data.
- Conducted **code reviews**, **mentored** junior developers, and ensured **best practices** in Python and front-end code, maintaining high standards across the codebase.

- Utilized **Agile** methodologies, including **Scrum**, to manage project deliverables and timelines, ensuring timely and high-quality software delivery.
- Diagnosed and resolved application issues in both backend and front-end environments, ensuring minimal disruption to healthcare operations.

Environment: Python, Django, Flask, FastAPI, PostgreSQL, MongoDB, HL7/FHIR, Pandas, Angular, React, JavaScript/TypeScript, HTML/CSS, Jenkins, Docker, AWS (Lambda, S3), Git, Agile/Scrum, HIPAA Compliance, FHIR Data Structures, pytest, NumPy

Python Developer (Volunteer)

Collx – United States | 10/2014 – 05/2016

- Developed and optimized **MySQL** failover scripts on **Linux**, ensuring seamless database failover during outages and maintaining data availability for critical applications.
- Designed and implemented **database models, APIs, and views** in Python to support interactive, web-based applications, enhancing data accessibility and user experience.
- Analyzed and fine-tuned **SQL queries** to improve performance, optimizing join costs for faster data retrieval.
- Integrated **Python with web development tools and web services**, enabling efficient backend processing and interaction with web applications.
- Wrote and executed **MySQL** database queries directly from Python using connectors like **Python-MySQL** and **MySQLdb**, supporting data operations and reporting needs.
- Created and maintained **unit tests using PyUnit** to ensure code reliability and functionality across Python applications, contributing to high-quality, bug-free software.
- Developed and tested **front-end components** for dashboards using Python, Bootstrap, CSS, and JavaScript, delivering a responsive and user-friendly interface.
- Conducted **statistical data analysis with Pandas**, supporting data-driven insights for financial applications.
- Leveraged **Python ORM libraries** such as **Django ORM** and **SQLAlchemy** to manage database interactions and streamline data manipulation.
- Troubleshooted, fixed, and deployed multiple Python bug fixes to **enhance application stability and performance**.
- Worked within **Agile** methodologies, participating in daily stand-ups, sprint planning, and code reviews to ensure efficient project delivery and team collaboration.

Environment: Python, Django, SQLAlchemy, MySQL, JavaScript, Bootstrap, Pandas, PyUnit, IDEs, Internal Debug Tools, Linux, Agile, Jenkins

Education

- **University of West Georgia | UWG**
Bachelor of Science in Computer Science | 08/2012 – 05/2016